15 June 2026

Heat Loss Report & System Design

Prepared for

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Prepared by

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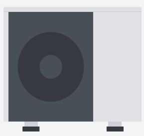
Summary

Your property's heat loss

Floor area	Total heat loss	Average heat load
206 m^2	**5.65** kW	**27** W/m^2

Proposed system design



HEAT PUMP

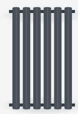
Vaillant aroTHERM pro 7kW

Capacity at 45°C flow temp and -1.5 °C outdoor air temperature	7.54 kW
SCOP at 45°C	4.13

UNDERFLOOR HEATING

63.2m^2 new underfloor

0.0m^2 retained



EMITTERS

9 new radiators

3 replaced, 6 additional, 2 retained

Performance comparison

Annual CO2 savings	Estimated annual running cost savings
2.9 to 3.2 tonnes	**£0 to £175** per year

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Heat loss report

This section presents the results of our detailed heat loss calculations for your property. The overall heat loss determines how big your heating system needs to be, and the room-by-room breakdown allows us to size the emitters (radiators and underfloor heating) correctly to keep each room warm on a cold day.

Introduction

How quickly your property loses heat depends on how big it is, the materials it is made from, and how air tight it is. Our detailed heat loss survey captured all this information and we have laid it out in this report so you can understand how and why we produce our recommendations. There are several key questions that we're looking to answer as we go through this process for you:

Are the heat pump and radiators big enough?

To keep your property warm, the heat pump needs to deliver heat to the emitters (by which we just mean radiators and underfloor heating) as quickly as your property loses it, and in turn the emitters need to deliver that heat as quickly as each room loses it.

We cannot rely on the fact that your existing radiators keep your property warm with your existing heating system, because heat pumps typically work at a much lower flow temperature than fossil-fuel boilers.

Whilst heat pumps are capable of making very hot water, they work most efficiently when generating low-temperature heat, typically between 35-50C. Gas boilers have traditionally been set up to generate high-temperature heat, with water temperatures of 70-80C (although they also work more efficiently at lower temperatures!).

A lower flow temperature, while more efficient, does mean the heat output from each radiator will be reduced. We use the heat loss survey to calculate how that reduced output compares to the room's demand for heat and then determine which (if any) radiators need replacing. In this way we ensure your new system will run efficiently with a low flow temperature, while still keeping you warm on cold days.

Is the heat pump too big?

This strikes many as a funny question, but we do not want to "oversize" your heat pump. While we design your heating system to ensure it keeps your property warm on cold days, most of the time it isn't that cold! On milder days, your house loses heat more slowly so the heat pump will need to provide less heat.

Heat pumps have to be sized quite precisely, as opposed to the majority of gas or oil boilers which are typically far more powerful than is actually needed for the property. If heat pumps turn on and off all the time, it can be really inefficient and unnecessarily increase energy bills. So we want to choose a model that can keep your property warm on a cold day, but "modulate down" to keep running efficiently on milder days.

We hope this all makes sense! Please do give the report a thorough read through and let us know if you have any questions – we're more than happy to explain what bits mean in more detail if you would like!

HEAT LOSS

Calculation conditions

When calculating the property's heat loss we design to certain conditions. This section shows the conditions used for this property.

Indoor & outdoor

Design outdoor air temperature

-1.5 °C

The "99th percentile" temperature for the area — the outdoor temperature only falls below this 1% of the time.

Indoor temperature

18 °C to 22 °C

The set point the system needs to maintain. The value used in each room is shown in the room by room section.

Design ground temperature

11 °C

Average ground temperature across the year, used to calculate heat loss through floors.

Ventilation & infiltration

Air permeability

11.0 $m^3/hr/m^2$ at 50 Pa

Source: Estimated from survey data

A measure of how much air leaks through the building fabric, expressed as the volume of air per hour per square metre of exposed surface area at a reference pressure of 50 Pa. This value is the starting point for calculating the ventilation heat loss in each room.

Shielding

Normal

An adjustment to account for how exposed the property is to the wind. Properties in sheltered locations lose less heat through ventilation than those in exposed locations.

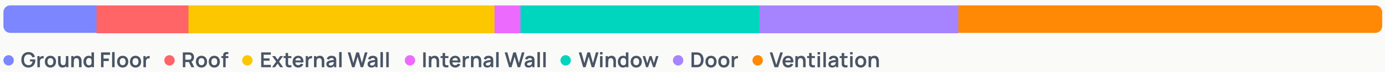
All calculations have been done in compliance with BS EN12831 (UK National Annex) and comply with the standards laid out in the Microgeneration Certification Scheme.

HEAT LOSS

Heat loss by element

This section shows the heat loss through each element in the property. They give you a sense of which parts of the property fabric lose the most heat, and may indicate areas where insulating could have a significant impact on the heat loss.

Total heat loss 5.65 kW



	U-VALUE* (W/m²K)	AREA	HEAT LOSS
Ground Floor 1. Extension 2023-onwards. Maximum value. 2. Solid floor. No insulation.	0.18 – 0.84	109.1 m²	352 W
Intermediate Floor/Ceiling • Intermediate floor. no insulation.	1.41 – 1.73	205.6 m²	0 W
Roof 1. Pitched, 150mm insulation at joists, felted 2. Extension 2023-onwards. Maximum value.	0.15 – 0.23	96.5 m²	375 W
External Wall 1. Filled Cavity. Brick and Standard Block (100mm). Plaster 2. Extension 2023-onwards. Maximum value.	0.18 – 0.45	184.4 m²	1,282 W
Internal Wall • Block - 100mm standard aerated block, plaster. 126mm.	1.06	283.2 m²	101 W
Window 1. Extension 2023-onwards. Maximum value. 2. Double Glazed, Wood/PVC frame	1.40 – 2.80	23.7 m²	979 W
Door 1. Extension 2023-onwards. Maximum value. 2. Wooden door. 50% double glazing.	1.40 – 2.80	22.6 m²	811 W

	ACH**	VOLUME	HEAT LOSS
Ventilation	0.46 – 1.04	493 m³	1,754 W
Total			5.65 kW

*** U-Value:** the thermal conductivity of the element.

****ACH:** air changes per hour

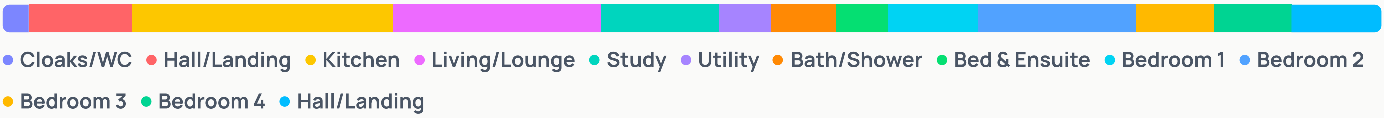
HEAT LOSS

Heat loss by room

This section shows the heat loss from each room in the property. These results are used to design the emitters (i.e. radiators or underfloor) for the new system.

Total heat loss

5.65 kW



	ROOM TEMP	ACH*	FLOOR AREA	VOLUME	HEAT LOSS	HEAT LOSS PER UNIT AREA
Cloaks/WC	18 °C	1.0	1.9 m²	5 m³	73 W	38 W/m²
Hall/Landing	18 °C	0.5	15.0 m²	36 m³	402 W	27 W/m²
Kitchen	18 °C	0.5	52.0 m²	125 m³	1,107 W	21 W/m²
Living/Lounge	21 °C	0.6	18.5 m²	44 m³	879 W	48 W/m²
Study	21 °C	0.5	12.5 m²	30 m³	505 W	40 W/m²
Utility	18 °C	0.5	9.3 m²	22 m³	187 W	20 W/m²
Bath/Shower	22 °C	0.5	5.2 m²	13 m³	238 W	46 W/m²
Bed & Ensuite	21 °C	0.6	7.5 m²	18 m³	225 W	30 W/m²
Bedroom 1	18 °C	0.6	19.2 m²	46 m³	381 W	20 W/m²
Bedroom 2	18 °C	0.6	16.8 m²	40 m³	665 W	40 W/m²
Bedroom 3	18 °C	0.5	15.6 m²	38 m³	283 W	18 W/m²
Bedroom 4	18 °C	0.5	13.3 m²	32 m³	319 W	24 W/m²
Hall/Landing	18 °C	0.5	18.8 m²	45 m³	390 W	21 W/m²

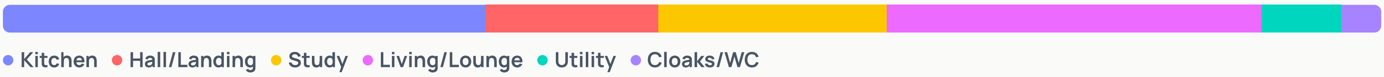
***ACH:** air changes per hour

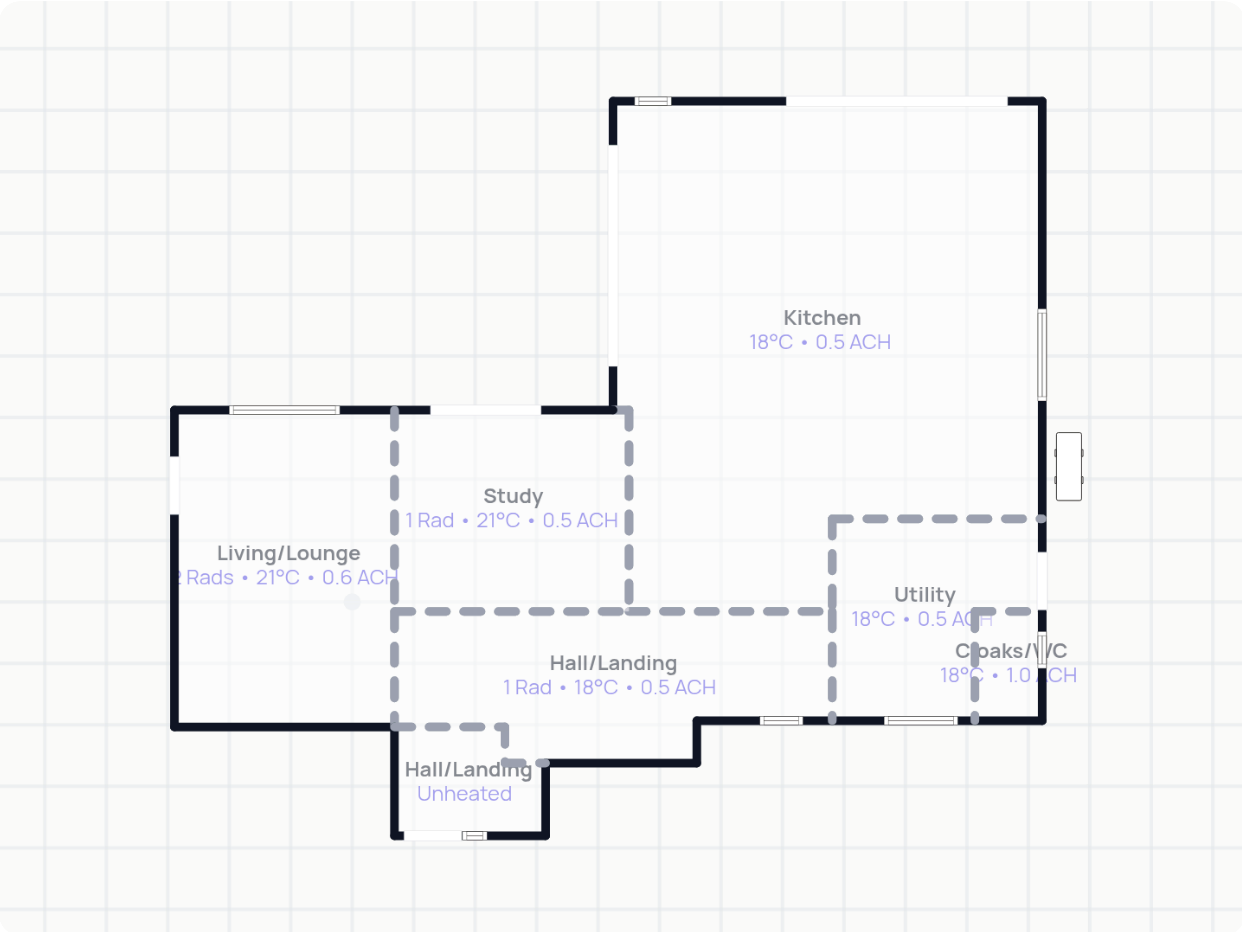
HEAT LOSS BY FLOOR

Ground floor

Heat loss by room

3153 W





Rads: number of radiators

UFH: underfloor heating in room

°C: room temperature

ACH: air changes per hour

W: number of windows

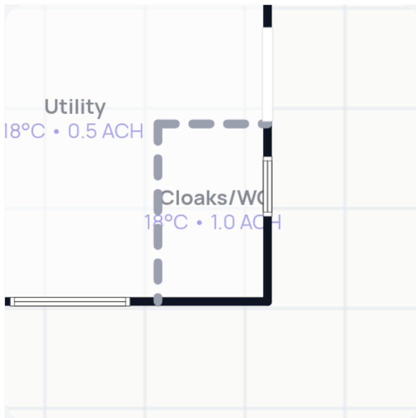
D: number of doors

HEAT LOSS BY ROOM

Cloaks/WC

Total heat loss	Room temp	Air changes per hour
73 W	18 °C	1.0

Floor area	Average height	Volume
1.9 m²	2.4 m	4.6 m³



Heat loss by element

73 W

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Ground Floor	11.0 °C	0.18 W/m²K	1.9 m²	2 W
Intermediate Ceiling	18.0 °C	1.73 W/m²K	1.9 m²	0 W
External Walls	-1.5 °C	0.18 W/m²K	6.2 m²	22 W
Internal Walls	18.0 °C	1.06 W/m²K	6.9 m²	0 W
Windows	-1.5 °C	1.40 W/m²K	0.6 m²	17 W
Ventilation	-1.5 °C			31 W

Heat loss per m² 38 W/m²

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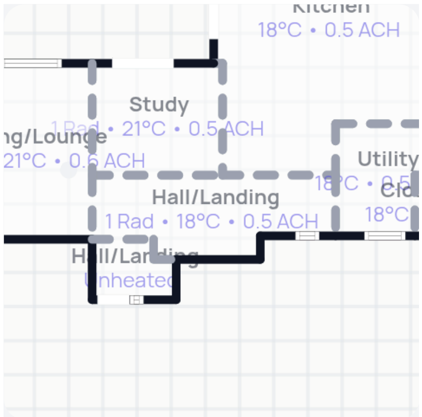
HEAT LOSS BY ROOM

Hall/Landing



Total heat loss	Room temp	Air changes per hour
402 W	18 °C	0.5

Floor area	Average height	Volume
15.0 m²	2.4 m	36.0 m³

Heat loss by element

402 W

• Ground Floor • External Walls • Internal Walls • Windows • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Ground Floor	11.0 °C	0.56 W/m²K	15.0 m²	59 W
Intermediate Ceiling	18.0 °C	1.73 W/m²K	15.0 m²	0 W
External Walls	-1.5 °C	0.45 W/m²K	11.8 m²	104 W
Internal Walls 1	18.0 °C	1.06 W/m²K	12.2 m²	0 W
Internal Walls 2	21.0 °C	1.06 W/m²K	13.7 m²	-43 W
Internal Walls 3	5.0 °C	1.06 W/m²K	7.3 m²	101 W
Windows	-1.5 °C	2.80 W/m²K	1.0 m²	57 W
Ventilation	-1.5 °C			126 W

Heat loss per m² 27 W/m²

Emitter performance

870 W (216%) of 402 W at 45 °C

• Radiator

EMITTER	IMAGE	DETAILS	EMITTER OUTPUT	% DEMAND MET*
Radiator		Type 22 (K2) 700 x 1200 mm	870 W	216 %

***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45 °C.

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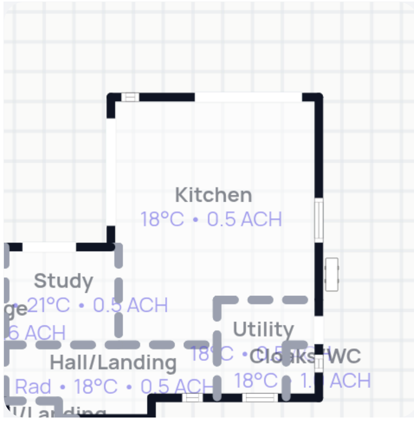
HEAT LOSS BY ROOM

Kitchen



Total heat loss **1107 W**	Room temp **18 °C**	Air changes per hour **0.5**
Floor area **52.0 m²**	Average height **2.4 m**	Volume **124.8 m³**

Heat loss by element

1,107 W

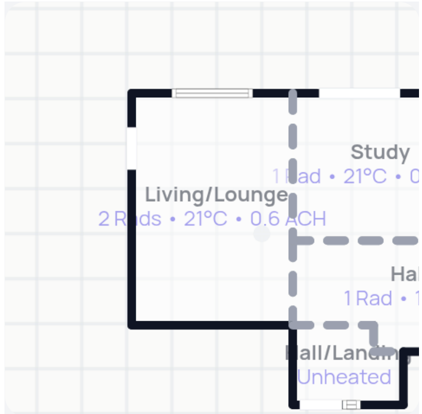
• Ground Floor • External Walls • Windows • External doors • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Ground Floor	11.0 °C	0.18 W/m²K	52.0 m²	65 W
Intermediate Ceiling	18.0 °C	1.73 W/m²K	52.0 m²	0 W
External Walls 1	-1.5 °C	0.45 W/m²K	14.7 m²	129 W
External Walls 2	-1.5 °C	0.18 W/m²K	12.4 m²	44 W
Internal Walls 1	21.0 °C	1.06 W/m²K	8.5 m²	-27 W
Internal Walls 2	18.0 °C	1.06 W/m²K	19.8 m²	0 W
Windows	-1.5 °C	1.40 W/m²K	2.8 m²	77 W
External doors	-1.5 °C	1.40 W/m²K	15.1 m²	413 W
Ventilation	-1.5 °C			405 W

Heat loss per m² 21 W/m²

HEAT LOSS BY ROOM

Living/Lounge



Total heat loss	Room temp	Air changes per hour
879 W	21 °C	0.6

Floor area	Average height	Volume
18.5 m²	2.4 m	44.4 m³

Heat loss by element

879 W

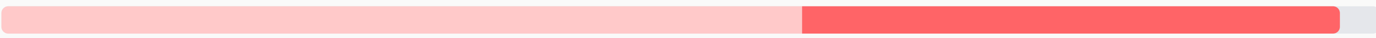
• Ground Floor • External Walls • Internal Walls • Windows • External doors • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Ground Floor	11.0 °C	0.84 W/m²K	18.5 m²	154 W
Intermediate Ceiling	21.0 °C	1.73 W/m²K	18.5 m²	0 W
External Walls	-1.5 °C	0.45 W/m²K	25.5 m²	259 W
Internal Walls 1	18.0 °C	1.06 W/m²K	4.5 m²	14 W
Internal Walls 2	21.0 °C	1.06 W/m²K	7.9 m²	0 W
Windows	-1.5 °C	2.80 W/m²K	2.2 m²	136 W
External doors	-1.5 °C	2.80 W/m²K	1.9 m²	117 W
Ventilation	-1.5 °C			198 W

Heat loss per m² **48 W/m²**

Emitter performance

852 W (97%) of 879 W at 45 °C



• Radiator • Radiator

EMITTER	IMAGE	DETAILS	EMITTER OUTPUT	% DEMAND MET*
Radiator		Type 21 (P+) 600 x 1200 mm	511 W	58 %
Radiator		Type 21 (P+) 600 x 800 mm	341 W	39 %

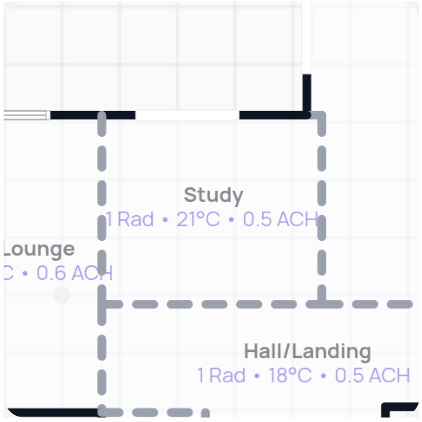
***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45 °C.

HEAT LOSS BY ROOM

Study

Total heat loss	Room temp	Air changes per hour
505 W	21 °C	0.5

Floor area	Average height	Volume
12.5 m²	2.4 m	30.0 m³



Heat loss by element

505 W

• Ground Floor • External Walls • Internal Walls • External doors • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Ground Floor	11.0 °C	0.48 W/m²K	12.5 m²	60 W
Intermediate Ceiling	21.0 °C	1.73 W/m²K	12.5 m²	0 W
External Walls	-1.5 °C	0.45 W/m²K	4.9 m²	50 W
Internal Walls 1	18.0 °C	1.06 W/m²K	17.6 m²	56 W
Internal Walls 2	21.0 °C	1.06 W/m²K	7.9 m²	0 W
External doors	-1.5 °C	2.80 W/m²K	3.6 m²	227 W
Ventilation	-1.5 °C			112 W

Heat loss per m² — 40 W/m²

Emitter performance

426 W (84%) of 505 W at 45 °C

• Radiator

EMITTER	IMAGE	DETAILS	EMITTER OUTPUT	% DEMAND MET*
Radiator		Type 21 (P+) 600 x 1000 mm	426 W	84 %

***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45 °C.

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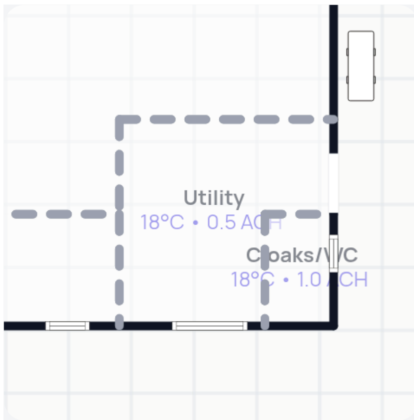
HEAT LOSS BY ROOM

Utility

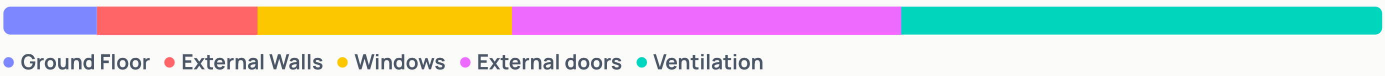
Total heat loss	Room temp	Air changes per hour
187 W	18 °C	0.5

Floor area	Average height	Volume
9.3 m²	2.4 m	22.3 m³



Heat loss by element

187 W



	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Ground Floor	11.0 °C	0.18 W/m²K	9.3 m²	12 W
Intermediate Ceiling	18.0 °C	1.73 W/m²K	9.3 m²	0 W
External Walls	-1.5 °C	0.18 W/m²K	6.0 m²	21 W
Internal Walls	18.0 °C	1.06 W/m²K	22.9 m²	0 W
Windows	-1.5 °C	1.40 W/m²K	1.3 m²	34 W
External doors	-1.5 °C	1.40 W/m²K	2.0 m²	54 W
Ventilation	-1.5 °C			66 W

Heat loss per m² 20 W/m²

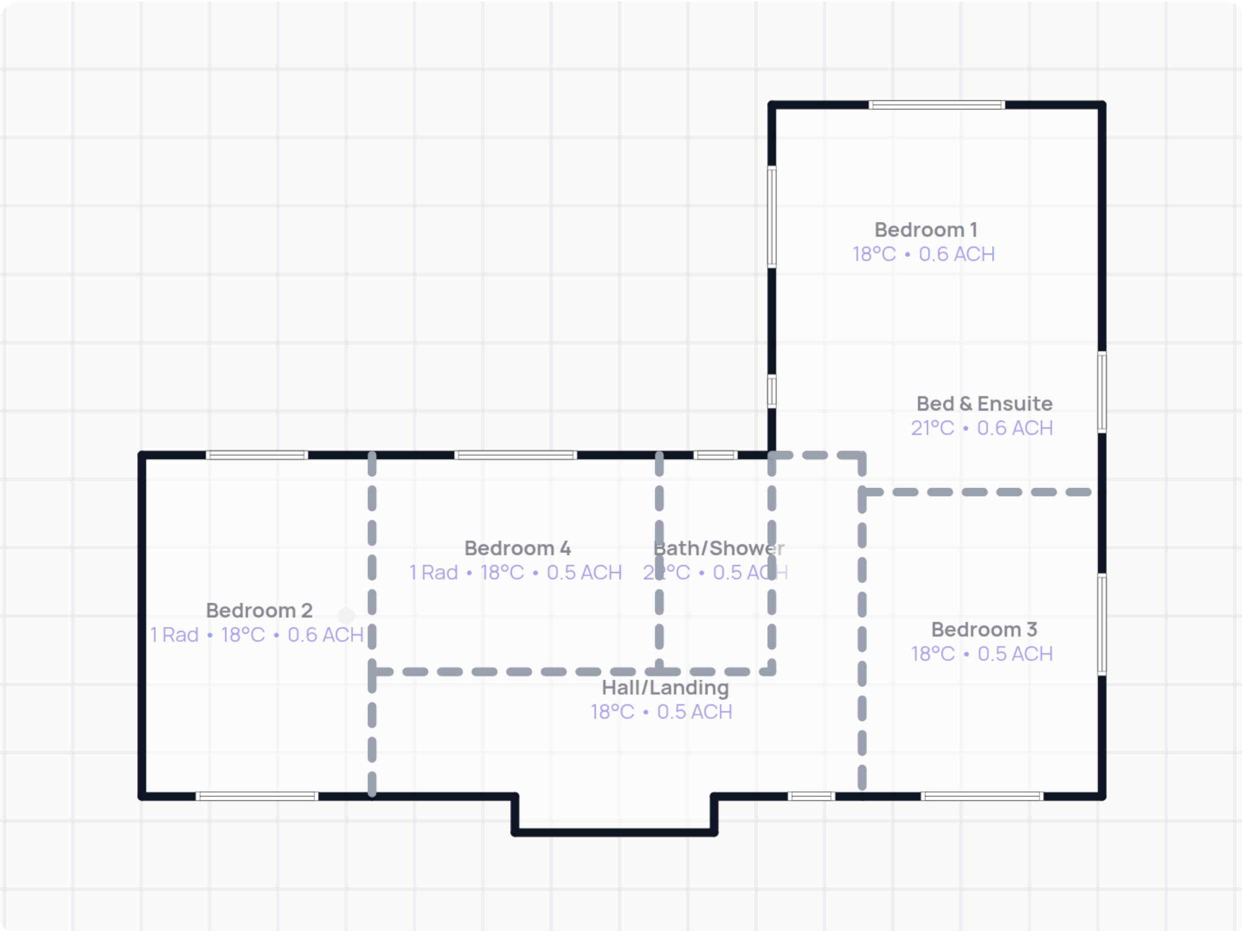
HEAT LOSS BY FLOOR

First floor

Heat loss by room

2501 W

• Bedroom 2 • Bedroom 4 • Bath/Shower • Hall/Landing • Bedroom 3 • Bed & Ensuite • Bedroom 1



Rads: number of radiators

UFH: underfloor heating in room

°C: room temperature

ACH: air changes per hour

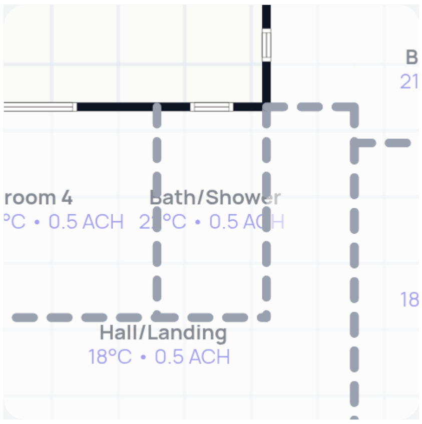
W: number of windows

D: number of doors

HEAT LOSS BY ROOM

Bath/Shower

Total heat loss 238 W	Room temp 22 °C	Air changes per hour 0.5
Floor area 5.2 m²	Average height 2.4 m	Volume 12.5 m³



Heat loss by element

238 W

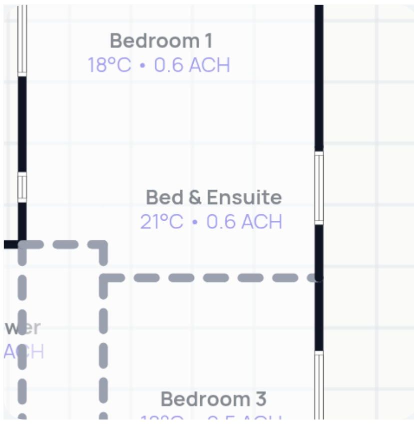
• Roof • External Walls • Internal Walls • Windows • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	22.0 °C	1.41 W/m²K	5.2 m²	0 W
Roof	-1.5 °C	0.23 W/m²K	5.2 m²	28 W
External Walls	-1.5 °C	0.45 W/m²K	3.3 m²	35 W
Internal Walls	18.0 °C	1.06 W/m²K	19.2 m²	81 W
Windows	-1.5 °C	2.80 W/m²K	0.7 m²	45 W
Ventilation	-1.5 °C			49 W

Heat loss per m² 46 W/m²

HEAT LOSS BY ROOM

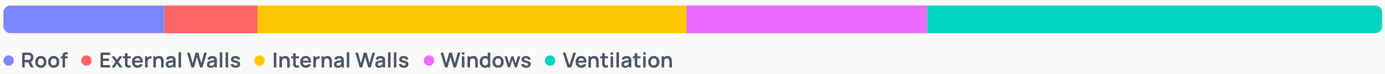
Bed & Ensuite



Total heat loss **225 W**	Room temp **21 °C**	Air changes per hour **0.6**
Floor area **7.5 m²**	Average height **2.4 m**	Volume **18.0 m³**

Heat loss by element

225 W



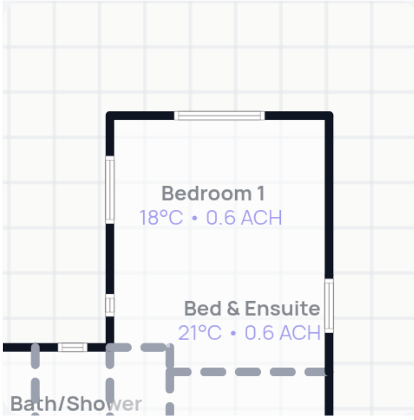
	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	21.0 °C	1.41 W/m²K	7.5 m²	0 W
Roof	-1.5 °C	0.15 W/m²K	7.5 m²	25 W
External Walls	-1.5 °C	0.18 W/m²K	3.9 m²	16 W
Internal Walls	18.0 °C	1.06 W/m²K	22.0 m²	70 W
Windows	-1.5 °C	1.40 W/m²K	1.3 m²	40 W
Ventilation	-1.5 °C			74 W

Heat loss per m² 30 W/m²

HEAT LOSS BY ROOM

Bedroom 1

Total heat loss **381 W**	Room temp **18 °C**	Air changes per hour **0.6**
Floor area **19.2 m²**	Average height **2.4 m**	Volume **46.1 m³**



Heat loss by element

381 W

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	18.0 °C	1.41 W/m²K	19.2 m²	0 W
Roof	-1.5 °C	0.15 W/m²K	19.2 m²	56 W
External Walls	-1.5 °C	0.18 W/m²K	28.8 m²	101 W
Internal Walls 1	21.0 °C	1.06 W/m²K	12.2 m²	-39 W
Internal Walls 2	18.0 °C	1.06 W/m²K	3.2 m²	0 W
Windows	-1.5 °C	1.40 W/m²K	3.6 m²	98 W
Ventilation	-1.5 °C			165 W

Heat loss per m² — 20 W/m²

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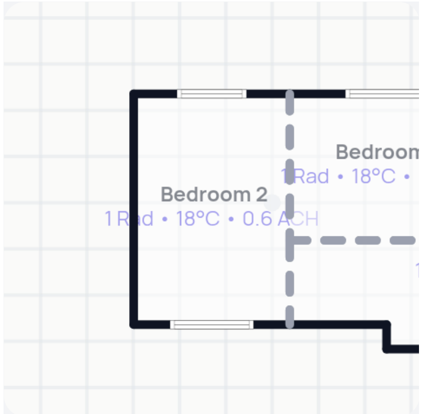
HEAT LOSS BY ROOM

Bedroom 2

Total heat loss **665 W**	Room temp **18 °C**	Air changes per hour **0.6**
Floor area **16.8 m²**	Average height **2.4 m**	Volume **40.3 m³**



Heat loss by element

665 W

• Roof • External Walls • Windows • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	18.0 °C	1.41 W/m²K	16.8 m²	0 W
Roof	-1.5 °C	0.23 W/m²K	16.8 m²	75 W
External Walls	-1.5 °C	0.45 W/m²K	24.2 m²	212 W
Internal Walls	18.0 °C	1.06 W/m²K	12.0 m²	0 W
Windows	-1.5 °C	2.80 W/m²K	4.0 m²	216 W
Ventilation	-1.5 °C			161 W

Heat loss per m² **40 W/m²**

Emitter performance

557 W (84%) of 665 W at 45 °C

• Radiator

EMITTER	IMAGE	DETAILS	EMITTER OUTPUT	% DEMAND MET*
Radiator		Type 21 (P+) 600 x 1100 mm	557 W	84 %

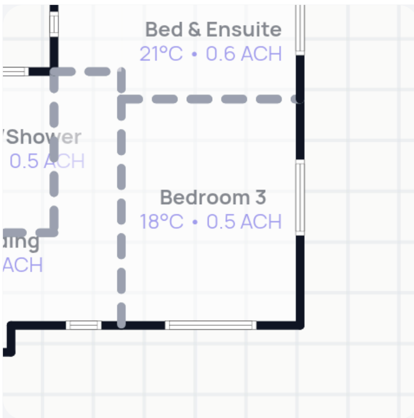
***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45 °C.

HEAT LOSS BY ROOM

Bedroom 3

Total heat loss	Room temp	Air changes per hour
283 W	18 °C	0.5

Floor area	Average height	Volume
15.6 m²	2.4 m	37.4 m³



Heat loss by element

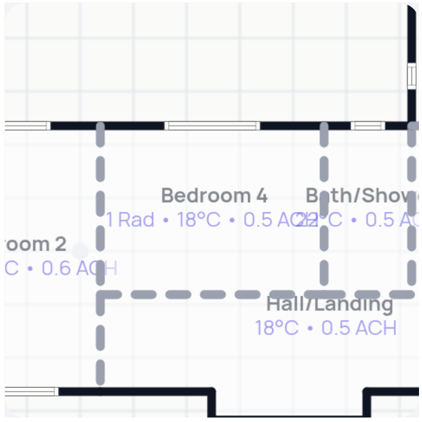
283 W

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	18.0 °C	1.41 W/m²K	15.6 m²	0 W
Roof	-1.5 °C	0.15 W/m²K	15.6 m²	46 W
External Walls	-1.5 °C	0.18 W/m²K	16.0 m²	56 W
Internal Walls 1	21.0 °C	1.06 W/m²K	8.4 m²	-27 W
Internal Walls 2	18.0 °C	1.06 W/m²K	10.7 m²	0 W
Windows	-1.5 °C	1.40 W/m²K	3.1 m²	84 W
Ventilation	-1.5 °C			124 W

Heat loss per m² — 18 W/m²

HEAT LOSS BY ROOM

Bedroom 4



Total heat loss	Room temp	Air changes per hour
319 W	18 °C	0.5

Floor area	Average height	Volume
13.3 m²	2.4 m	31.9 m³

Heat loss by element

319 W

• Roof • External Walls • Windows • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	18.0 °C	1.41 W/m²K	13.3 m²	0 W
Roof	-1.5 °C	0.23 W/m²K	13.3 m²	60 W
External Walls	-1.5 °C	0.45 W/m²K	7.9 m²	69 W
Internal Walls 1	18.0 °C	1.06 W/m²K	17.7 m²	0 W
Internal Walls 2	22.0 °C	1.06 W/m²K	7.6 m²	-32 W
Windows	-1.5 °C	2.80 W/m²K	2.2 m²	118 W
Ventilation	-1.5 °C			104 W

Heat loss per m² **24 W/m²**

Emitter performance

304 W (95%) of 319 W at 45 °C

• Radiator

EMITTER	IMAGE	DETAILS	EMITTER OUTPUT	% DEMAND MET*
Radiator		Type 21 (P+) 600 x 600 mm	304 W	95 %

***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45 °C.

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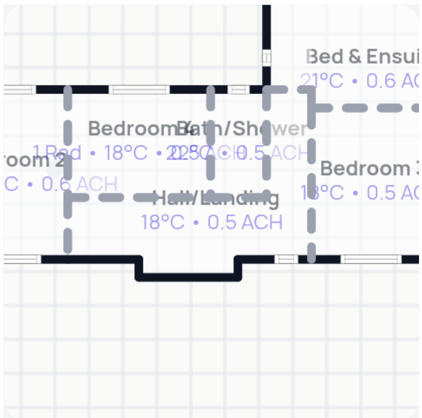
HEAT LOSS BY ROOM

Hall/Landing



Total heat loss	Room temp	Air changes per hour
390 W	18 °C	0.5

Floor area	Average height	Volume
18.8 m²	2.4 m	45.1 m³

Heat loss by element

390 W

• Roof • External Walls • Windows • Ventilation

	OTHER SIDE TEMPERATURE	U-VALUE	AREA	HEAT LOSS
Intermediate Floor	18.0 °C	1.41 W/m²K	18.8 m²	0 W
Roof	-1.5 °C	0.23 W/m²K	18.8 m²	84 W
External Walls	-1.5 °C	0.45 W/m²K	18.7 m²	164 W
Internal Walls 1	18.0 °C	1.06 W/m²K	28.3 m²	0 W
Internal Walls 2	21.0 °C	1.06 W/m²K	1.3 m²	-4 W
Internal Walls 3	22.0 °C	1.06 W/m²K	11.6 m²	-49 W
Windows	-1.5 °C	2.80 W/m²K	1.0 m²	57 W
Ventilation	-1.5 °C			138 W

Heat loss per m² 21 W/m²

HEAT LOSS

Ventilation heat loss breakdown

This section shows how much heat escapes with warm air leaving the building. This is calculated from the building's air permeability — how draughty it is overall — then distributed across rooms by their exposed surface area.

Whole home permeability

At 50 Pa	At normal conditions
11.0 m³/hr/m²	**0.55** m³/hr/m²

This figure is estimated based on the following contribution from each part of the property (each at 50 Pa, in m³/hr/m²):

Structural Masonry	+ 7.0
Floor Sealed	+ 0.0
Doors & windows 75% draught-proofed	+ 3.0
Storeys 2 storeys	+ 1.0
Total	= 11.0

Calculation factors

Shielding — **Normal**

How exposed the property is to the wind. Sheltered locations lose less heat through ventilation than exposed ones.

Zone height — **Between 5m and 10m**

The building's total height across all floors, used to determine the shielding factor.

Shielding factor (50Pa to normal conditions conversion) — **0.05**

We multiply 50 Pa permeability by this factor to derive the value at normal conditions.

	EXPOSED AREA	INFILTRATION EX. VENTS	VENTS	INFILTRATION FROM VENTS	TOTAL INFILTRATION	VOLUME	ACH*
Kitchen	97 m²	53 m³/hr	–	–	53 m³/hr	125 m³	0.50 /hr
Hall/Landing	35 m²	19 m³/hr	–	–	19 m³/hr	36 m³	0.54 /hr
Study	21 m²	12 m³/hr	–	–	12 m³/hr	30 m³	0.50 /hr
Living/Lounge	48 m²	26 m³/hr	–	–	26 m³/hr	44 m³	0.60 /hr
Utility	18 m²	10 m³/hr	–	–	10 m³/hr	22 m³	0.46 /hr
Cloaks/WC	9 m²	5 m³/hr	–	–	5 m³/hr	5 m³	1.04 /hr
Bedroom 2	45 m²	25 m³/hr	–	–	25 m³/hr	40 m³	0.61 /hr
Bedroom 4	23 m²	13 m³/hr	–	–	13 m³/hr	32 m³	0.50 /hr
Bath/Shower	9 m²	5 m³/hr	–	–	5 m³/hr	13 m³	0.50 /hr
Hall/Landing	39 m²	21 m³/hr	–	–	21 m³/hr	45 m³	0.47 /hr

	EXPOSED AREA	INFILTRATION EX. VENTS	VENTS	INFILTRATION FROM VENTS	TOTAL INFILTRATION	VOLUME	ACH*
Bedroom 3	35 m^2	19 m^3/hr	—	—	19 m^3/hr	38 m^3	0.51 /hr
Bed & Ensuite + Bedroom 1	64 m^2	35 m^3/hr	—	—	35 m^3/hr	64 m^3	0.55 /hr

***ACH:** the number of air changes per hour that the room sees. This is calculated based on the infiltration values.

System design

This section presents our proposed system design for the property. The design is based on the detailed room by room calculations in the previous section.

Proposed heat pump

Based on the property's design heat loss of 5.65kW and on all the attributes of the property that we know of at this stage, we would suggest the following heat pump.

Vaillant aroTHERM pro 7kW	
Capacity at 45 °C (-1.5 °C)	7.54 kW
SCOP at 45 °C	4.13
Flow temperature	45 °C

This heat pump covers **133 %** of the heating requirement at and above the design temperature of **-1.5 °C**. The heat pump will be run on weather compensation so the efficiency will be higher when the weather is milder.

MCS certificate number
011-1W1213_01

Refrigerant
R290

Model number
VWL 75/7.1 A 230V S2

Sound power
49 dB(A)

Width
1104 mm

Height
750 mm

Depth
454 mm

Weight
91 kg

Capacity (kW)

OUTSIDE TEMP	35°C	40°C	45°C	50°C	55°C
-5°C	6.51	6.65	6.78	6.31	5.84
-3°C	6.87	7.02	7.16	6.69	6.22
0°C	7.73	7.83	7.92	7.31	6.69
2°C	8.45	8.47	8.49	7.8	7.1
5°C	9.68	9.41	9.15	8.25	7.35

Seasonal Coefficient of Performance (SCOP)

SCOP is the average coefficient of performance over the heating season, accounting for the variation in outdoor temperature. Manufacturers have to test their equipment in standard conditions and report their SCOP performance data to be eligible for certification.

	35°	40°	45°	50°	55°
SCOP	4.74	4.43	4.13	3.82	3.52

SYSTEM DESIGN

Emitters

Emitters (i.e., radiators/fan coils/underfloor heating) are a vital part of the heating system. They take the heat produced by the heat source and distribute that heat around the property.

Based on the room by room heat loss results, we propose the following emitter design for the property. This design will ensure each room can be heated to its set point when it's -1.5 °C outside whilst maintaining high system efficiency.

Design conditions

- Flow temperature = 45 °C
- Temperature drop across the radiator (delta T) = 5 °C
- Mean radiator temperature = 42.5 °C
- UFH mean water temperatures (MWT):
  - Manifold 1 = 42.5 °C

Emitter replacements

We propose replacing or adding emitters in the following rooms:

1. Study
2. Living/Lounge
3. Bedroom 2
4. Bedroom 4
5. Bath/Shower
6. Hall/Landing
7. Bedroom 3
8. Bed & Ensuite
9. Bedroom 1

SYSTEM DESIGN

Current emitters in the property

ROOM	IMAGE	TYPE	DETAILS	EMITTER OUTPUT	ROOM OUTPUT	ROOM DEMAND	% DEMAND MET*
Kitchen		–		0 W	0 W	1107 W	0 %
Hall/Landing		Radiator	Type 22 (K2) 700 x 1200 mm	870 W	870 W	402 W	216 %
Study		Radiator	Type 21 (P+) 600 x 1000 mm	426 W	426 W	505 W	84 %
Living/Lounge		Radiator	Type 21 (P+) 600 x 1200 mm	511 W	852 W	879 W	97 %
		Radiator	Type 21 (P+) 600 x 800 mm	341 W			
Utility		–		0 W	0 W	187 W	0 %
Cloaks/WC		–		0 W	0 W	73 W	0 %
Bedroom 2		Radiator	Type 21 (P+) 600 x 1100 mm	557 W	557 W	665 W	84 %
Bedroom 4		Radiator	Type 21 (P+) 600 x 600 mm	304 W	304 W	319 W	95 %
Bath/Shower		–		0 W	0 W	238 W	0 %
Hall/Landing		–		0 W	0 W	390 W	0 %
Bedroom 3		–		0 W	0 W	283 W	0 %
Bed & Ensuite + Bedroom 1		–		0 W	0 W	606 W	0 %
		–		0 W			

***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45.0 °C.

SYSTEM DESIGN

Proposed emitter changes

ROOM	STATUS	IMAGE	TYPE	DETAILS	EMITTER OUTPUT	ROOM DEMAND	% DEMAND MET*
Kitchen	New		UFH	Solid (Screed) - 100% covered. Ceramic tiles, 5mm (0W/m²K) 150mm centres 42.5°C MWT	7294 W	1107 W	659 %
Hall/Landing	Keep		Radiator	Type 22 (K2) 700 x 1200 mm	870 W	402 W	216 %
Study	New		Radiator	Stelrad · Type 22 (K2) · White 1600 x 500 mm	567 W	505 W	112 %
	Remove		~~Radiator~~	~~Type 21 (P+)~~ ~~600 x 1000 mm~~	~~426 W~~		
Living/Lounge	Replacement		Radiator	Type 21 (p+) - White 600 x 1200 mm	611 W	879 W	108 %
	Keep		Radiator	Type 21 (P+) 600 x 800 mm	341 W		
Utility	New		UFH	Solid (Screed) - 100% covered. Ceramic tiles, 5mm (0W/m²K) 150mm centres 42.5°C MWT	1301 W	187 W	696 %
Cloaks/WC	New		UFH	Solid (Screed) - 100% covered. Ceramic tiles, 5mm (0W/m²K) 150mm centres 42.5°C MWT	273 W	73 W	374 %
Bedroom 2	Replacement		Radiator	Type 22 (K2) - White 600 x 1100 mm	730 W	665 W	110 %
Bedroom 4	Replacement		Radiator	Type 22 (K2) - White 600 x 600 mm	398 W	319 W	125 %
Bath/Shower	New		Radiator	Stelrad · Curved · White 1744 x 500 mm	265 W	238 W	111 %
Hall/Landing	New		Radiator	Type 22 (K2) - White 500 x 800 mm	460 W	390 W	118 %
Bedroom 3	New		Radiator	Type 21 (p+) - White 600 x 700 mm	362 W	283 W	128 %
Bed & Ensuite + Bedroom 1	New		Radiator	Stelrad · Straight · White 1211 x 500 mm	182 W	606 W	115 %
	New		Radiator	Type 21 (p+) - White 600 x 1000 mm	517 W		

***% demand met:** This is calculated for a day when the outdoor temperature is -1.5 °C and the flow temperature is 45.0 °C.

Sound assessment

Incomplete

To class as "permitted development" the heat pump design must comply with regulations regarding the sound level at a neighbour's nearest window/door. This section presents the results of the sound assessment we've conducted for the proposed heat pump and location, as set out in MCS 020a v1.0

This assessment accounts for:

- The sound level of the heat pump itself
- The influence of the space that it is in
- Any barriers between the heat pump and the assessment position

Results by position

Position 1	0 (Incomplete

SOUND ASSESSMENT

Position 1

Sound pressure level

Sound power level Vaillant aroTHERM pro 7kW VWL 75/7.1 A 230V S2	49 dB(A)
Reduction due to distance Distance (heat pump to assessment position): 0 m	0 dB(A)
Reduction due to barriers null	0 dB(A)
Total	**0 dB(A)**

Result

Incomplete

Maximum allowed value	37 dB(A)
Final result at assessment position	**0 dB(A)**

Performance estimate

Predicting running costs with a heat pump is hard as it depends on a lot of factors, many of which are outside our control. But it's clearly really important that you're able to make an informed decision. So in this section we give you our best estimate of what the running costs and carbon savings will look like for the property if and when you switch to a heat pump.

We then provide an in-depth description of how we've calculated these figures in case you want to dig into the details. Please do ask us if anything doesn't make sense!

Summary

Bills

Current	£942 to £1,043
Proposed system	£776 to £1,043
Savings	£0 to £175

Carbon

Current	3.4 to 3.8 tonnes CO_2
Proposed system	0.5 to 0.6 tonnes CO_2
Savings	85%

PERFORMANCE ESTIMATE

Detailed results

The running costs and carbon emissions of a heating system depends on a few key factors, all of which are explained in (even) more detail further down this report:

1. **Your property's heating and hot water demand**
   This is how much heat your property needs for heating and hot water in the year irrespective of what heating system is used to provide it. It's surprisingly hard to put a definitive number on this because it depends on how often you have the heating on and at what temperature, how much hot water you use, and the weather outside.

   Given this uncertainty we have estimated this in 2 different ways to try and give you the best understanding:
   - **EPC** – Based on the heating and hot water demand from the property's EPC. This assumes the property hasn't changed since the EPC date and uses assumptions about the heating and hot water system that don't apply to a heat pump. If the property has changed since the last EPC or that EPC had errors, this might not be the best measure, but we have to include it.
   - **Heating Degree Days** – Based on the heat loss calculations we have done combined with "heating degree day" data which represents the typical annual weather pattern in the area.

2. **The efficiency of your system**
   The efficiency of your system describes how much heat your system makes from what you pay for (electricity/gas/oil), both in heating and when making hot water. This depends on the system design, install quality, and how you run the system. We've included two sets of numbers here. One using fixed efficiencies from MCS and another using the Manufacturer's tested values.

   The MCS estimate uses standard efficiency values, which are the same for all heat pumps and only depend on the flow temperature the system is designed at. They tend to be more conservative. Manufacturer's efficiencies use their test data at standard conditions. We've included both to give you a fair picture of what to expect.

3. **Energy prices**
   Heat pump running costs depend on the price of electricity, and the comparison with your old system depends on the price of mains gas. We've included numbers both at the price cap and with a heat pump tariff, to show the range of what you might pay depending on the tariff you are on. We've included more info on heat pump tariffs further down.

4. **Emission factors**
   Emission factors show how much carbon dioxide is emitted per unit of energy provided by a fuel. They are a measure of how "clean" a fuel is.

PERFORMANCE ESTIMATE

MCS031 Performance estimate

The table below shows a performance estimate in MCS's required table layout following the methodology laid out in MCS 031 Issue 4.0.

Heat Pump System Performance Estimate	
Your energy requirements	
Energy required for heating	**12,165** kWh
Demand to be supplied by the heat pump	**12,165** kWh
Energy required for hot water	**2,938** kWh
Demand to be supplied by the heat pump	**2,938** kWh
Your property	
Your postcode prefix	**EX**
Total property floorspace (not property footprint)	**206** m²
Average watts per square metre Note: W/m2 is a measure of your property's thermal efficiency. 0-30W/m2 is very low heat loss and 120-150W/m2 is very high heat loss.	**27** W/m²
Proposed system	
Heat pump capacity	**7.5** kW
Heat pump type	**ASHP**
Your system is proposed to provide:	**Space heating only**
Your proposed heating system will be (select one):	**Upgraded radiators**
The proposed flow temperature will be	**45** °C
Performance	
The Seasonal Performance Factor is calculated to be:	**3.4**
Estimate of energy consumption of the proposed heat pump (or combined system where Hybrid). Note: you can convert these figures to approximate running costs.	**High: 5,837** kWh → **Low: 4,775** kWh
Important Note: This is not a detailed system design. It offers a reasonable estimate of likely performance and a description of the likely design. This estimate is based on a full heat loss survey and design.	
Applicable warning notes (from Table 2): None	

EcoSphere Energy

PERFORMANCE ESTIMATE

Results table - Manufacturer's efficiencies

This table shows a granular breakdown of the potential energy consumption and running costs of your heat pump, based on different estimates of your property's energy demand and different energy tariffs that you might choose. We are also using heat pump efficiencies (i.e. SCOP) based on the manufacturer's test data, which often shows higher efficiencies than MCS's fixed values.

Warning: In accordance with MCS standards you should treat any performance estimate that differs from the MCS031 calculations with caution. We have included it because we think it's helpful to show you how different estimates of your heating demand affect the overall performance, how the tariff you choose can affect the running costs of the system, and the potential efficiency of the heat pump based on the manufacturer's own test data.

If you have any questions about this, please do let us know and we'll be happy to talk you through it.

	BASED ON EPC		BASED ON HEAT LOSS CALCULATIONS AND HEATING DEGREE DAYS		BASED ON LAST YEARS CONSUMPTION
Bills	Existing (Mains gas)*	Heat pump (Electricity)	Existing (Mains gas)*	Heat pump (Electricity)	Unavailable
Price cap (24.67 p/kWh)	£942	£933 £9	£1,043	£1,043 £0	
Octopus Cosy (20.53 p/kWh)	£942	£776 £166	£1,043	£868 £175	
Your Tariff (24.67 p/kWh)	£942	£933 £9	£1,043	£1,043 £0	

* Based on an existing Mains gas price of 5.74 p/kWh

Carbon	Existing (Mains gas)	Heat pump (Electricity)	Existing (Mains gas)	Heat pump (Electricity)	Unavailable
Totals	3,447 kg	514 kg 2,933 kg	3,814 kg	575 kg 3,239 kg	

Consumption	Existing (Mains gas)	Heat pump (Electricity)	Existing (Mains gas)	Heat pump (Electricity)	Unavailable
Heating	13,223 kWh	2,946 kWh	13,417 kWh	2,989 kWh	
Hot Water	3,193 kWh	835 kWh	4,746 kWh	1,240 kWh	
Total	16,416 kWh	3,780 kWh	18,164 kWh	4,229 kWh	

Demand					Unavailable
Heating	12,165 kWh		12,344 kWh		
Hot Water	2,938 kWh		4,366 kWh		
Total	15,103 kWh		16,710 kWh		

Efficiency	Existing (Mains gas)	Heat pump (Electricity)	Existing (Mains gas)	Heat pump (Electricity)	Unavailable
Heating	92 %	413 %	92 %	413 %	
Hot water	92 %	352 %	92 %	352 %	

PERFORMANCE ESTIMATE

Inputs and assumptions

In each of the following sections, we break down how we've calculated the numbers that appear in the results table.

Heat energy required

To work out how much energy the property needs to keep it warm, we first have to work out how much heat is required. That might sound strange, but because different heating sources have wildly different efficiencies, this heat energy number will need to be divided by the system efficiency (see next section) to calculate the actual energy required.

	BASED ON EPC	BASED ON HEAT LOSS CALCULATIONS AND HEATING DEGREE DAYS	BASED ON LAST YEARS CONSUMPTION
	Certificate number: 2876-1120-1937-4111-1136	1858 degree days, 277 W/°C	Unavailable
Heating	12,165 kWh	12,344 kWh	
Hot water	2,938 kWh	4,366 kWh	
Total	15,103 kWh	16,710 kWh	

System efficiency

The efficiency of a heating system describes how much useful output (heat) you get per unit of what you pay for (electricity/gas/oil). A system's efficiency will often be different when it is doing space heating vs. when it's making hot water because the temperature that the system is heating the water to is often hotter in water heating cycles. We have used the following inputs when modelling the system.

Existing heating system

Existing heating system fuel:	Mains gas
Efficiency - space heating:	92%
Efficiency - hot water:	92%

Heat pump system

Type of system:	Air source heat pump
Model:	Vaillant aroTHERM pro 7kW
MCS certificate numbers:	011-1W1213_01
Flow temperature - space heating:	45°C
Flow temperature - hot water:	55°C

	MCS SPF	MANUFACTURERS SCOP
Heating efficiency	340 %	413 %
Hot water efficiency	170 %	352 %

Energy prices

As the last few years have shown, energy prices are very hard to predict. In modelling bills we have used the current energy price cap and also shown you a few different heat pump tariff options. Energy tariffs can also vary by supplier and change over time, so we'd recommend not making your decision based purely on a specific tariff or running cost figure.

There are an increasing number of heat pump tariffs that give you access to cheaper electricity. These are typically **time of use** tariffs which charge you different amounts depending on when you use the electricity. So you can set the hot water heating schedule (and to a lesser extent the space heating schedule) to make use of lower prices.

If you have solar (or are planning to install it), the electricity prices will also be lower because the heat pump will use some of the electricity that the solar is generating. This is especially true for hot water consumption because you can schedule the hot water to run in the middle of the day.

MAINS GAS	
Existing price	5.74 p/kWh

ELECTRICITY	
Price cap Price cap 1st April - 30th June 2026	24.67 p/kWh
Octopus Cosy Based on Octopus's expectations of customer consumption patterns on the Cosy Octopus tariff as of April 2026	20.53 p/kWh

Emission factors

We've used the emission factors from the government's SAP 2010 methodology. For mains gas this value is fixed but for electricity it is falling over time as the grid decarbonises. This means the system will get cleaner and cleaner. The value used here is based on a projected continued reduction in grid carbon intensity.

Mains gas

210 gCO2/kWh

Electricity

136 gCO2/kWh

Dependence of performance on flow temperature

How much electricity the heat pump will use to provide heating depends on the flow temperature that the system runs at. We have designed the system to run at 45°C when it's -1.5°C outside and we'll set it up to use weather compensation so it runs more efficiently at milder temperatures.

To demonstrate the importance of flow temperature, the table below shows how much electricity the system would consume to provide heating at a range of flow temperatures. This graph is based on the heat loss calculations and heating degree days based estimate, but the pattern would be similar for any of the inputs.

	35°	40°	45°	50°	55°
SCOP	4.74	4.43	4.13	3.82	3.52
Electricity consumed (kWh/year)	3525	3772	4046	4374	4747

PERFORMANCE ESTIMATE

MCS Key facts - Energy Performance Estimate

Predicting the heat demand of a building, and therefore the performance and running costs of heating systems, is difficult to predict with certainty due to the variables discussed here. These variables apply to all types of heating systems, although the efficiency of heat pumps is more sensitive to good system design and installation. For these reasons your estimate is given as guidance only and should not be considered as a guarantee.

Seasonal Coefficient of Performance:

MCS Seasonal Coefficient of Performance (SCoP) is derived from the EU ErP labelling requirements, and is a theoretical indication of the anticipated efficiency of a heat pump over a whole year using standard (i.e. not local) climate data for 3 locations in Europe. It is used to compare the relative performance of heat pumps under fixed conditions and indicates the units of total heat energy generated (output) for each unit of electricity consumed (input). As a guide, a heat pump with a MCS SCoP of 3 indicates that 3 kWh of heat energy would be generated for every 1 kWh of electrical energy it consumes over a 'standard' annual cycle.

Energy Performance Estimate

An Energy Performance Certificate (EPC) is produced in accordance with a methodology approved by the government. As with all such calculations, it relies on the accuracy of the information input. Some of this information, such as the insulating and air tightness properties of the building may have to be assumed and this can affect the final figures significantly leading to uncertainty especially with irregular or unusual buildings.

Identifying the uncertainties of energy predictions for heating systems

We have identified 3 key types of factor that can affect how much energy a heating system will consume and how much energy it will deliver into a property. These are 'Fixed', 'Variable' and 'Random'. Most factors are common to ALL heating systems regardless of the type (e.g oil, gas, solid fuel, heat pump etc.) although the degree of effect varies between different types of heating system as given in the following table.

The combined effect of these factors on energy consumption and the running costs makes overall predictions difficult however an accuracy ± 25-30% would not be unreasonable in many instances. Under some conditions even this could be exceeded (e.g. considerable opening of windows). Therefore it is advised that when making choices based on mainly financial criteria (e.g. payback based on capital cost versus net benefits such as fuel savings and financial incentives) this variability is taken into account as it could extend paybacks well beyond the period of any incentives received, intended occupancy period, finance agreement period etc.

'Fixed' which include:

Factor	Impact
Equipment Selection Performance figures (SCoP) from ErP data	System Efficiency
Energy Assessment via the EPC (e.g. assumptions as to fabric construction and levels of insulation; the variation in knowledge and experience of Energy Assessors)	Energy Required

'Variable' which are affected by the system design and include:

Factor	Impact
Accuracy of sizing of heat pump-i.e. closeness of unit output selection (kW) to demand heat requirement (kW)	System Efficiency
Design space and ambient (external) temperatures	Energy Required
Design flow / return water temperatures and weather compensation	System Efficiency
Type of Heat emitter (e.g. Under-floor; natural convector (e.g. radiator), fan convector etc.)	System Efficiency

'Random' which cannot be anticipated and include:

Factor	Impact
User behaviour:	
• Room temperature settings	Energy Required
• Hot water usage and temperature settings	Energy Required
• Occupancy patterns/times	Energy Required
• Changing the design HP flow temperatures	System Efficiency
• Ventilation (i.e. opening windows)	Energy Required
Annual climatic variations (i.e. warmer and colder years than average)	Energy Required

Key

The statement at the end of each item indicates the major factor affected as follows:

Energy Required: the heat energy output requirement of the system which directly impacts on running costs. This requirement exists regardless of the heating system chosen as it is the heat required to keep the space comfortable. Opening windows or increasing room temperatures will demand more heat output, which means more energy input but this would NOT directly affect the efficiency. Thus increased energy demand does NOT automatically mean reduced efficiency.

System Efficiency: the efficiency of the system has been directly affected and will therefore demand more input energy to achieve the same heat output thus increasing running costs. However, increased energy input does NOT necessarily mean lower system efficiency (see above).